12TH CHEMISTRY · CHAPTER 26 · FULL CHAPTER MCQS

Polymers

135 topic-wise MCQs from the complete chapter with answers and reasons

INTRODUCTION

1. Very large molecules made by joining many small molecules are called:

(a) polymers (b) monomers
(c) isomers (d) synthons

Answer: (a) polymers — The small molecules are the monomers.

2. The small molecules which join to form a polymer are called:

(a) plasticizers (b) polymers
(c) monomers (d) initiators

Answer: (c) monomers — Many monomers join by polymerisation.

3. Shopping bags are commonly made of:

(a) Kevlar (b) poly(ethene)
(c) nylon-6 (d) PMMA

Answer: (b) poly(ethene) — A light and soft polymer.

4. Clothesline ropes are made of:

(a) PET (b) Kevlar
(c) PVC (d) poly(propene)

Answer: (d) poly(propene) — Given in the book's introduction.

5. Bulletproof vests are made from:

(a) poly(propene) (b) poly(ethene)
(c) Kevlar (d) PVC

Answer: (c) Kevlar — Kevlar is hard and thermally very stable.

6. The systematic name of polythene is:

(a) poly(ethene) (b) poly(ethane)
(c) poly(ethyne) (d) poly(chloroethene)

Answer: (a) poly(ethene) — The prefix poly is written before the monomer name.

7. The systematic name of polyvinyl chloride is:

(a) poly(propene) (b) poly(chloroethane)
(c) poly(ethene) (d) poly(chloroethene)

Answer: (d) poly(chloroethene) — Its monomer is chloroethene.

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8. PETE stands for:

(a) polyethene tetraester (b) polyethylene terephthalate
(c) polyester ethene (d) polyethyl ether

Answer: (b) polyethylene terephthalate — The common name of the polyester.

9. Common names are used for polymers with two or more monomers because the systematic name is:

(a) too long (b) unknown
(c) incorrect (d) the same for all

Answer: (a) too long — E.g. poly(ethane-1,2-diyl benzene-1,4-dicarboxylate).

10. The properties of a polymer depend on the type and number of monomers and on the:

(a) shape of the container (b) colour of the monomer
(c) price of the monomer (d) method of manufacture

Answer: (d) method of manufacture — Stated in the introduction.

26.1 ADDITION POLYMERISATION

11. The process of joining monomers to form a polymer is called:

(a) cracking (b) hydrolysis
(c) polymerisation (d) distillation

Answer: (c) polymerisation — Definition.

12. The number of main types of polymerisation is:

(a) two (b) three
(c) four (d) five

Answer: (a) two — Addition and condensation.

13. In addition polymerisation, the atoms of the monomers:

(a) are partly lost as water (b) all become part of the polymer
(c) are lost as HCl (d) form a salt

Answer: (b) all become part of the polymer — Nothing is lost.

14. Monomers for addition polymerisation must contain:

(a) an amino group (b) two –OH groups
(c) two –COOH groups (d) a C=C double bond

Answer: (d) a C=C double bond — The double bond opens up to link the monomers.

15. During addition polymerisation the side product is:

(a) hydrogen chloride (b) water
(c) none (d) ammonia

Answer: (c) none — No small molecule is produced.

16. Addition polymerisation often takes place through a:

(a) neutralisation (b) chain reaction
(c) step-by-step condensation (d) hydrolysis

Answer: (b) chain reaction — Started by an initiator.

17. An initiator starts the chain reaction by forming:

(a) free radicals (b) water
(c) salts (d) esters

Answer: (a) free radicals — It is a very reactive molecule.

18. The monomer of poly(ethene) is:

(a) $CH_3–CH=CH_2$
(b) $CH_2=CHCl$
(c) $CH_3–CH_3$
(d) $CH_2=CH_2$

Answer: (d) $CH_2=CH_2$ — Ethene.

19. The repeat unit of poly(ethene) is:

(a) $–CH_2–CHCl–$
(b) $–CH_2–CH_2–$
(c) –CH=CH–
(d) $–CH_2–CH(CH_3)–$

Answer: (b) $–CH_2–CH_2–$ — The double bond becomes a single bond.

20. In poly(ethene), the value of n can be up to about:

(a) 50 (b) 10
(c) 100 (d) 10 000

Answer: (d) 10 000 — n is very large.

21. The chains of LDPE are:

(a) more branched (b) linear
(c) cross-linked by esters (d) ring shaped

Answer: (a) more branched — Branching prevents close packing.

22. The chains of HDPE are:

(a) cyclic (b) highly branched
(c) linear (d) ionic

Answer: (c) linear — They pack closely.

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23. LDPE is used for:

(a) shopping bags and light plastic bottles
(b) bulletproof vests
(c) artificial corneas
(d) gears and bearings

Answer: (a) shopping bags and light plastic bottles — It is very light and soft.

24. HDPE is produced using:

(a) caprolactam
(b) concentrated sulfuric acid
(c) a plasticizer
(d) a Ziegler–Natta catalyst

Answer: (d) a Ziegler–Natta catalyst — At low pressure and temperature.

25. HDPE is tough and dense because its chains:

(a) cannot pack tightly
(b) pack closely
(c) contain chlorine
(d) contain amide links

Answer: (b) pack closely — Linear chains give a compact structure.

26. The Ziegler–Natta catalyst was initially discovered by Karl Ziegler in:

(a) 1972
(b) 1903
(c) 1953
(d) 1990

Answer: (c) 1953 — From the Did You Know box.

27. Giulio Natta used the catalyst family to make:

(a) stereoregular polymers
(b) nylon-6,6
(c) plasticizers
(d) Kevlar

Answer: (a) stereoregular polymers — E.g. crystalline polypropylene.

28. The monomer of PVC is:

(a) ethene
(b) chloroethene
(c) propene
(d) chloroethane

Answer: (b) chloroethene — CH_2=CHCl.

29. The repeat unit of PVC is:

(a) –CH_2–CCl_2–
(b) –CH_2–CH_2–
(c) –CHCl–CHCl–
(d) –CH_2–CHCl–

Answer: (d) –CH_2–CHCl– — From chloroethene.

30. PVC is formed by:

(a) hydrolysis
(b) condensation polymerisation
(c) addition polymerisation
(d) esterification

Answer: (c) addition polymerisation — Its monomer has a C=C bond.

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31. Chemicals added to make a material softer and more flexible are called:

(a) plasticizers (b) initiators
(c) catalysts (d) monomers

Answer: (a) plasticizers — Definition.

32. Plasticizers work by:

(a) forming cross links
(b) reducing the forces between polymer chains
(c) increasing chain length
(d) removing chlorine

Answer: (b) reducing the forces between polymer chains — The chains can then move more freely.

33. Rigid PVC is:

(a) a liquid (b) soft and flexible
(c) hard and stiff (d) a good conductor

Answer: (c) hard and stiff — It contains no plasticizer.

34. PVC is self-extinguishing because of its:

(a) plasticizers (b) hydrogen atoms
(c) double bonds (d) chlorine atoms

Answer: (d) chlorine atoms — They make it resistant to burning.

35. PVC is a good:

(a) electrical insulator (b) electrical conductor
(c) fuel (d) oxidising agent

Answer: (a) electrical insulator — So it is used on wires and cables.

36. PVC pipes are used for water supply because of their:

(a) low melting point (b) high conductivity
(c) corrosion resistance and durability (d) solubility in water

Answer: (c) corrosion resistance and durability — Given under applications.

37. Which of these is a use of PVC?

(a) bulletproof vests (b) blood bags and catheters
(c) stockings (d) skateboard wheels

Answer: (b) blood bags and catheters — Tubing, blood bags and catheters.

38. Rigid PVC does not react with most:

(a) monomers (b) plasticizers
(c) initiators (d) acids, bases or oils

Answer: (d) acids, bases or oils — It is chemically inert.

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26.1 CONDENSATION POLYMERISATION

39. In condensation polymerisation, each time a new bond forms:

(a) nothing is lost
(b) a double bond opens
(c) a free radical forms
(d) a small molecule is eliminated

Answer: (d) a small molecule is eliminated — Usually water.

40. The small molecule eliminated in condensation polymerisation is usually:

(a) hydrogen
(b) oxygen
(c) water
(d) carbon dioxide

Answer: (c) water — It can also be HCl, CH_3OH or NH_3.

41. Monomers for condensation polymerisation must have at least:

(a) one double bond
(b) two reactive functional groups
(c) one triple bond
(d) one benzene ring

Answer: (b) two reactive functional groups — On different ends of the monomer.

42. Condensation polymerisation typically takes place in a:

(a) step-by-step manner
(b) chain reaction
(c) single step
(d) free-radical explosion

Answer: (a) step-by-step manner — As shown in the book's figure.

43. Polyesters contain the linkage:

(a) –C–C– only
(b) –CO–NH–
(c) –CO–O–
(d) –O–O–

Answer: (c) –CO–O– — Ester link.

44. A molecule with two –OH groups is called a:

(a) diamine
(b) diol
(c) dioyl chloride
(d) diene

Answer: (b) diol — E.g. ethane-1,2-diol.

45. Terylene (PET) is made from ethane-1,2-diol and:

(a) benzene-1,4-dicarboxylic acid
(b) hexanedioic acid
(c) 1,6-diaminohexane
(d) chloroethene

Answer: (a) benzene-1,4-dicarboxylic acid — A diol and a dicarboxylic acid.

46. When a diol reacts with a dicarboxylic acid, the molecule removed is:

(a) methanol
(b) hydrogen chloride
(c) ammonia
(d) water

Answer: (d) water — –OH reacts with –COOH.

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47. Polyesters are suitable for making fibres because of their:

(a) low melting point (b) high tensile strength
(c) solubility in water (d) high conductivity

Answer: (b) high tensile strength — Stated in the book.

48. Polyester is used as an alternative to cotton because of its low cost and ability to:

(a) conduct heat (b) absorb water
(c) resist creases (d) dissolve in oil

Answer: (c) resist creases — Stated in the book.

49. Plastic bottles for beverages are made from:

(a) PMMA (b) nylon-6
(c) Kevlar (d) PETE

Answer: (d) PETE — A polyester.

50. Glass fibre reinforced plastics contain cross-linked:

(a) polyester (b) poly(ethene)
(c) PVC (d) nylon-6

Answer: (a) polyester — Polyester is used as cross links to obtain hard plastics.

51. Polyamides contain the linkage:

(a) –C=C– (b) –CO–O–
(c) –CO–NH– (d) –O–H

Answer: (c) –CO–NH– — Amide link.

52. A molecule with two $-NH_2$ groups is called a:

(a) diamine (b) diol
(c) dicarboxylic acid (d) dioyl chloride

Answer: (a) diamine — E.g. 1,6-diaminohexane.

53. A dioyl chloride contains two:

(a) –OH groups (b) –COCl groups
(c) $-NH_2$ groups (d) –Cl atoms on a ring

Answer: (b) –COCl groups — It is more reactive than a dicarboxylic acid.

54. Compared with dicarboxylic acids, dioyl chlorides are:

(a) unreactive (b) less reactive
(c) equally reactive (d) more reactive

Answer: (d) more reactive — So they are used to make polyamides easily.

55. An amino acid contains the groups:

(a) –OH and –COOH (b) $-NH_2$ and –COOH
(c) $-NH_2$ and –OH (d) –COCl and –OH

Answer: (b) $-NH_2$ and –COOH — Both groups are in one molecule.

56. The bond formed between two amino acids is a:

(a) peptide bond (b) ester bond
(c) glycosidic bond (d) hydrogen bond

Answer: (a) peptide bond — A type of amide linkage.

57. Proteins are natural:

(a) poly(alkenes) (b) polyesters
(c) polyamides (d) plasticizers

Answer: (c) polyamides — Made from amino acids.

58. An acyl chloride reacts with an amine to give an amide and:

(a) Cl_2 (b) H_2O
(c) NH_3 (d) HCl

Answer: (d) HCl — R–COCl + R'–NH_2 → R–CO–NH–R' + HCl.

59. The reaction of an acyl chloride with ammonia takes place at:

(a) room temperature (b) very high temperature
(c) very high pressure only (d) in the presence of UV light

Answer: (a) room temperature — Acyl chlorides are fairly reactive.

60. Ethanoyl chloride reacts with ammonia to form:

(a) ethyl ethanoate (b) ethylamine
(c) ethanoic acid (d) ethanamide

Answer: (d) ethanamide — $CH_3COCl + NH_3 \rightarrow CH_3CONH_2 + HCl$.

26.2 NYLON

61. Nylon is a synthetic:

(a) polysaccharide (b) polyester
(c) poly(alkene) (d) polyamide

Answer: (d) polyamide — It has amide links.

62. Nylon-6,6 is made from 1,6-diaminohexane and:

(a) benzene-1,4-dicarboxylic acid (b) ethane-1,2-diol
(c) hexanedioic acid (d) caprolactam

Answer: (c) hexanedioic acid — Adipic acid.

63. The common name of hexanedioic acid is:

(a) terephthalic acid (b) adipic acid
(c) lactic acid (d) acetic acid

Answer: (b) adipic acid — Used with hexamethylenediamine.

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64. The small molecule removed in the formation of nylon-6,6 is:

(a) water (b) hydrogen chloride
(c) ammonia (d) methanol

Answer: (a) water — From –NH2 and –COOH.

65. Nylon-6 is formed from:

(a) ethene (b) adipic acid
(c) caprolactam (d) chloroethene

Answer: (c) caprolactam — A single cyclic monomer.

66. Caprolactam has:

(a) a straight chain only (b) a cyclic structure
(c) two benzene rings (d) no nitrogen

Answer: (b) a cyclic structure — A ring containing an amide group.

67. Nylon-6 is so named because its monomer has:

(a) six carbon atoms (b) six nitrogen atoms
(c) six rings (d) six oxygen atoms

Answer: (a) six carbon atoms — Caprolactam has six carbons.

68. During the formation of nylon-6:

(a) ammonia is removed (b) water is removed
(c) HCl is removed (d) no small molecule is removed

Answer: (d) no small molecule is removed — Yet it is classed as a condensation polymer.

69. Nylon-6 is classified as a condensation polymer because of the:

(a) loss of water
(b) mode of formation of the amide linkage
(c) presence of C=C bonds
(d) use of an initiator

Answer: (b) mode of formation of the amide linkage — Stated in the book.

70. Compared with nylon-6,6, nylon-6 has a:

(a) rougher surface (b) higher melting point
(c) lower melting point (d) lower impact resistance

Answer: (c) lower melting point — So it is easier to mould.

71. Nylon-6 is mostly used in:

(a) blood bags (b) shopping bags
(c) cling films (d) rollers and skateboard wheels

Answer: (d) rollers and skateboard wheels — To avoid shocks.

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72. The monomers of nylon-6,10 are hexane-1,6-diamine and:

(a) decanedioic acid (b) hexanedioic acid
(c) butanedioic acid (d) ethanedioic acid

Answer: (a) decanedioic acid — The second number shows ten carbons in the acid.

73. The diacid monomer of nylon-6,4 has:

(a) ten carbon atoms (b) six carbon atoms
(c) four carbon atoms (d) two carbon atoms

Answer: (c) four carbon atoms — $HOOC(CH_2)_2COOH$.

74. Nylon has a slippery surface which:

(a) reduces friction (b) increases friction
(c) absorbs water (d) conducts electricity

Answer: (a) reduces friction — A listed property.

75. Nylon is generally resistant to:

(a) all acids (b) oils, many solvents and alkalis
(c) heat only (d) nothing

Answer: (b) oils, many solvents and alkalis — A listed property.

76. The melting point of nylon is:

(a) zero (b) very low
(c) below room temperature (d) high

Answer: (d) high — A listed property.

77. Nylon is used for ropes and fishing nets because of its:

(a) low strength
(b) high tensile strength and resistance to water wear
(c) solubility in water
(d) brittleness

Answer: (b) high tensile strength and resistance to water wear — Stated under applications.

78. Gears and bearings are made of nylon because of its:

(a) strength, low friction and heat resistance
(b) softness
(c) low melting point
(d) transparency

Answer: (a) strength, low friction and heat resistance — Automotive uses.

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79. Which of these is a consumer use of nylon?

(a) water pipes (b) blood bags
(c) zippers and toothbrushes (d) cling film

Answer: (c) zippers and toothbrushes — Listed in the book.

80. Nylon is stretchable due to its:

(a) density (b) brittleness
(c) rigidity (d) elasticity

Answer: (d) elasticity — A listed property.

26.3–26.4 IDENTIFYING POLYMERS AND MONOMERS

81. If the main chain of a polymer has only carbon atoms, it is:

(a) a polyamide (b) a polyester
(c) an addition polymer (d) a condensation polymer

Answer: (c) an addition polymer — Formed from alkenes.

82. If the main chain contains O or N atoms with carbon, the polymer is:

(a) an alkane (b) an addition polymer
(c) a poly(alkene) (d) a condensation polymer

Answer: (d) a condensation polymer — Polyester or polyamide.

83. A C=O group directly next to an oxygen in the chain shows:

(a) an amide linkage (b) an ester linkage
(c) an ether linkage (d) a peptide linkage

Answer: (b) an ester linkage — The polymer is a polyester.

84. A C=O group directly next to an N–H in the chain shows:

(a) an amide linkage (b) an ester linkage
(c) an ether linkage (d) a glycosidic linkage

Answer: (a) an amide linkage — The polymer is a polyamide.

85. PTFE is:

(a) a polyamide (b) a polyester
(c) an addition polymer (d) a natural polymer

Answer: (c) an addition polymer — Poly(tetrafluoroethene) has only C–C bonds in its chain.

86. PLA, poly(lactic acid), is a:

(a) polyester (b) polyamide
(c) poly(alkene) (d) protein

Answer: (a) polyester — It has –COO– links.

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87. Kevlar is a:

(a) polyester (b) polyamide
(c) poly(alkene) (d) polysaccharide

Answer: (b) polyamide — It has –CONH– links.

88. Polymers containing benzene rings in the chain are called:

(a) natural polymers (b) aliphatic polymers
(c) addition polymers only (d) aromatic polymers

Answer: (d) aromatic polymers — E.g. PET or Kevlar.

89. The smallest section that repeats itself in a polymer is the:

(a) initiator (b) monomer
(c) repeat unit (d) plasticizer

Answer: (c) repeat unit — It is written in brackets with subscript n.

90. To find the monomer of an addition polymer, the C–C bond of the repeat unit is changed into:

(a) a C–O bond (b) a C=C double bond
(c) a C–N bond (d) a C≡N bond

Answer: (b) a C=C double bond — This gives back the alkene.

91. To find the monomers of a condensation polymer:

(a) water is added back (b) hydrogen is removed
(c) chlorine is added (d) the chain is oxidised

Answer: (a) water is added back — The ester or amide bond is broken.

92. When an amide bond is broken to find the monomers, –OH is added to the:

(a) chain end only (b) N side
(c) ring (d) C=O side

Answer: (d) C=O side — –H is added to the N side.

93. The monomers of Kevlar are benzene-1,4-dicarboxylic acid and:

(a) ethane-1,2-diol (b) benzene-1,4-diamine
(c) 1,6-diaminohexane (d) phenylethene

Answer: (b) benzene-1,4-diamine — Found by breaking the amide bonds.

94. Monomers containing C=C bonds undergo:

(a) saponification (b) condensation polymerisation
(c) hydrolysis (d) addition polymerisation

Answer: (d) addition polymerisation — E.g. ethene, propene, phenylethene.

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95. Another name for phenylethene is:

(a) styrene (b) toluene
(c) xylene (d) aniline

Answer: (a) styrene — Given in the book.

96. A polymer made from two different monomers is called a:

(a) dimer (b) homopolymer
(c) copolymer (d) plasticizer

Answer: (c) copolymer — E.g. from ethene and propenoic acid.

97. An example of a single monomer carrying two different functional groups is:

(a) lactic acid (b) ethene
(c) adipic acid (d) chloroethene

Answer: (a) lactic acid — It has –OH and –COOH.

98. Carboxylic groups and alcohols condense to give:

(a) a polyester and HCl (b) a polyamide and H_2O
(c) a polyester and H_2O (d) a polyamide and HCl

Answer: (c) a polyester and H_2O — Listed in the book.

99. Acyl chlorides and alcohols condense to give:

(a) a polyamide and H_2O (b) a polyester and H_2O
(c) a polyamide and HCl (d) a polyester and HCl

Answer: (d) a polyester and HCl — Listed in the book.

100. Amino groups and acyl chlorides condense to give:

(a) a polyamide and H_2O (b) a polyamide and HCl
(c) a polyester and HCl (d) a polyester and H_2O

Answer: (b) a polyamide and HCl — Listed in the book.

101. The polymer with repeat unit –CH_2–CH(C_6H_5)– is made from:

(a) chloroethene (b) propene
(c) benzene (d) phenylethene

Answer: (d) phenylethene — Change C–C back to C=C.

102. The formula of propenoic acid is:

(a) CH_3–COOH
(b) CH_3–CH_2–COOH
(c) H_2C=CH–COOH
(d) HOOC–COOH

Answer: (c) H_2C=CH–COOH — It forms a copolymer with ethene.

26.5 NON-DEGRADABLE AND DEGRADABLE POLYMERS

103. The most widely used polymeric material on the globe is:

(a) nylon-6
(b) plastic such as poly(ethene)
(c) Kevlar
(d) PMMA

Answer: (b) plastic such as poly(ethene) — Stated in the book.

104. Non-degradable polymers:

(a) cannot be broken down into simpler non-toxic compounds
(b) dissolve in water
(c) are decomposed by bacteria
(d) break down in light

Answer: (a) cannot be broken down into simpler non-toxic compounds — So they pollute the environment.

105. Tiny pieces formed when large plastic pieces break down are called:

(a) plasticizers
(b) monomers
(c) microplastics
(d) micelles

Answer: (c) microplastics — They pollute soil, water and air.

106. Microplastics are harmful because they can:

(a) dissolve harmlessly
(b) biodegrade quickly
(c) act as fertilisers
(d) enter the food chain

Answer: (d) enter the food chain — They may harm human health.

107. Marine animals are harmed by plastic waste because they:

(a) use it for shelter only
(b) digest it easily
(c) mistake it for food
(d) convert it to energy

Answer: (c) mistake it for food — It causes choking, injury or starvation.

108. Non-degradable plastics take up huge space in:

(a) the atmosphere
(b) landfills
(c) the ozone layer
(d) factories

Answer: (b) landfills — They do not decompose.

109. A plastic that is decomposed by microorganisms when buried is called:

(a) biodegradable
(b) photodegradable
(c) non-degradable
(d) thermosetting

Answer: (a) biodegradable — Definition.

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110. Plastics containing starch are degraded faster because:

(a) starch is a plasticizer
(b) starch absorbs UV light
(c) starch is inert
(d) bacteria attack natural polymers easily

Answer: (d) bacteria attack natural polymers easily — Stated in the book.

111. Soil organisms break down polyesters and polyamides through:

(a) hydrolysis (b) oxidation by ozone
(c) addition (d) sublimation

Answer: (a) hydrolysis — Ester and amide links are hydrolysable.

112. Hydrolysis of a polyamide produces carboxylic acids and:

(a) alcohols (b) amines
(c) alkenes (d) ethers

Answer: (b) amines — From the amide link.

113. Hydrolysis of a polyester produces carboxylic acids and:

(a) alkenes (b) amines
(c) alcohols (d) ketones

Answer: (c) alcohols — From the ester link.

114. A polymer which can enhance the biodegradability of a plastic is:

(a) HDPE (b) PVC
(c) PTFE (d) PLA

Answer: (d) PLA — Poly(lactic acid).

115. Polymers degraded by light are called:

(a) inert (b) biodegradable
(c) photodegradable (d) thermoplastic

Answer: (c) photodegradable — They absorb UV light.

116. The group built into chains to make polymers photodegradable is:

(a) C–F (b) C–H
(c) C–C (d) C=O

Answer: (d) C=O — Carbonyl groups have enhanced photosorption.

117. Photodegradation is caused by absorption of:

(a) infrared light (b) UV light
(c) radio waves (d) sound

Answer: (b) UV light — It weakens bonds in and between chains.

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118. A drawback of photodegradable plastics is that in landfills:

(a) no light is available (b) too much light is present
(c) they burn (d) they dissolve

Answer: (a) no light is available — So they do not degrade there.

119. Recycling of photodegradable polymers is:

(a) impossible to attempt (b) very efficient
(c) always required (d) not efficient

Answer: (d) not efficient — Stated in the book.

120. Dissolvable stitches are made from:

(a) poly(alkenes) (b) polyesters
(c) PVC (d) PTFE

Answer: (b) polyesters — They slowly break down in the body by hydrolysis.

121. Poly(alkenes) are difficult to biodegrade because they are:

(a) soluble in water (b) very reactive
(c) chemically inert (d) natural polymers

Answer: (c) chemically inert — They have strong C–C and C–H bonds.

122. Alkaline hydrolysis of a polyamide gives an amine and a:

(a) carboxylate salt (b) carboxylic acid
(c) alcohol (d) ketone

Answer: (a) carboxylate salt — In alkali the acid forms its salt.

123. Acidic hydrolysis of a polyester gives:

(a) a carboxylic acid and an amine (b) a carboxylate salt and an alcohol
(c) a carboxylic acid and an alcohol (d) an alkene and water

Answer: (c) a carboxylic acid and an alcohol — No salt is formed in acid.

26.6 POLYMERS IN ARTIFICIAL ORGANS

124. Man-made devices that replace or support damaged body parts are called:

(a) plasticizers (b) artificial organs
(c) catalysts (d) copolymers

Answer: (b) artificial organs — Definition.

125. A polymer that does not harm the body or cause rejection is said to be:

(a) photodegradable (b) biodegradable
(c) biocompatible (d) volatile

Answer: (c) biocompatible — A required property.

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126. Artificial blood vessels are made from PTFE and:

(a) PET (b) PVC
(c) PMMA (d) nylon-6

Answer: (a) PET — Smooth surfaces prevent blood clots.

127. Artificial heart valves are made from specialised polyurethanes and:

(a) starch (b) collagen
(c) PMMA (d) silicone

Answer: (d) silicone — Highly flexible and durable.

128. Artificial skin uses collagen and:

(a) PTFE (b) PVC
(c) polylactic acid (PLA) (d) UHMWPE

Answer: (c) polylactic acid (PLA) — It acts as a scaffold for new skin growth.

129. Artificial hip and knee joints use:

(a) PMMA (b) UHMWPE
(c) PLA (d) collagen

Answer: (b) UHMWPE — Ultra-high molecular weight polyethylene.

130. The artificial cornea is made from hydrogels and:

(a) UHMWPE (b) PTFE
(c) PET (d) PMMA

Answer: (d) PMMA — Poly(methyl methacrylate).

131. The polymer for an artificial cornea must be:

(a) perfectly clear (b) opaque
(c) coloured (d) conducting

Answer: (a) perfectly clear — To restore vision.

132. UHMWPE in artificial joints provides:

(a) a scaffold for skin (b) a rough surface
(c) a transparent window (d) a smooth, low-friction surface

Answer: (d) a smooth, low-friction surface — It allows movement and resists wear.

133. Collagen used in artificial skin is a:

(a) synthetic polyester (b) natural polymer
(c) poly(alkene) (d) plasticizer

Answer: (b) natural polymer — PLA is the synthetic one.

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134. Polymers for blood vessels must be flexible, whereas those for bones must be:

(a) strong (b) soft
(c) liquid (d) soluble

Answer: (a) strong — Properties are tuned to the body part.

135. Materials for heart valves must be resistant to:

(a) sound (b) light
(c) clotting (d) magnetism

Answer: (c) clotting — Stated in Table 26.2.

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